Tianyi Cai

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Summary

Profile
- Applied scientist with 9 years of experience combining causal inference, machine learning, experimentation, forecasting, and scalable engineering to drive strategic decision-making.

Skills
- Causal inference: doubly robust estimators, quasi-experimental methods, heterogeneous treatment effects.
- Machine learning: tabular/panel/time series data, temporal feature engineering, model selection and validation.
- Experimentation: study design, power and minimum detectable effect, bandit algorithms.
- Forecasting: anomaly detection, backtesting and calibration, scenario analysis.
- Engineering: CI/CD pipelines, distributed computing, model deployment and monitoring.
- Leadership: cross-functional collaboration, thought leadership, stakeholder communication and management.

Toolkit
- Python, SQL, R, Anyscale/Ray, MLflow, Airflow, Snowflake, Databricks, Claude Code.

Experience

Staff Data Scientist, Research, Consumer Trading — **Google**, New York, NY 08/2026–Present
- Technical lead for Search Discover Data Science

Staff Applied Scientist, Consumer Trading — **Coinbase**, New York, NY 01/2025–05/2026
- Architected agentic workflows for causal inference using custom skills/MCPs; scaled incrementality analyses from weeks to hours, shaping product go/no-go decisions and incentive sizing (Coinbase One Card, new product/asset cannibalization).
- Deployed machine learning models predicting customer LTV (long-term experimentation metric), trading volume, and churn (high-net-worth targeting); outputs informed strategy across marketing, sales, and CX.
- Owned the end-to-end forecasting system (calibrated and interpretable production models) for 10+ key metrics; replaced subjective and manual weekly leadership reporting with automated anomaly detection and scenario analysis workflows.
- Engineered unified temporal data pipelines powering forecasting, machine learning, and causal inference models across the broader consumer organization, leveraging daily data snapshots to ensure point-in-time correctness and prevent data leakage.

Senior Data Scientist, Planner Evaluation — **Waymo**, New York, NY 11/2022–01/2025
- Halved the family-wise error rate in onboard software evaluation by applying confidence interval adjustments.
- Pioneered the use of statistical power and minimum detectable effect as evaluation metrics throughout the Planner Evaluation team by authoring the API and supporting documentation.
- Provided statistical mentorship and workshops for teams across Waymo on topics such as resampling methods, Type I/II error, and quantifying uncertainty.

Senior Research Data Scientist, Quantitative Engineering — **Meta**, New York, NY 02/2021–11/2022
- Doubled user engagement with infrastructure tools by replacing unreliable front-end North Star metrics with robust back-end metrics and identifying their causal drivers, in collaboration with engineering, product, design, and UX research.
- Segmented developer workflows by fine-tuning a pre-trained transformer model to extract high-dimensional vector embeddings for computational notebooks; directly informed the engineering roadmap for user-level custom features.
- Audited the experimentation strategy for measuring p99 latency in query services; designed a formal statistical framework incorporating variance reduction and alternative metrics.

Applied Scientist II, Global Intelligence — **Uber**, San Francisco, CA 01/2020–02/2021
- Quantified uncertainty in competitive intelligence metrics, providing statistically rigorous signals that underpinned executive decisions on global market exits and consolidations.
- Ensembled machine learning models using internal and third-party data to identify rare user groups for multimillion-dollar experiments; improved the legacy model by doubling average precision.

Senior Data Scientist, Data and Research — **BitSight**, Boston, MA 10/2017–01/2020
- Led all data science projects for third-party cybersecurity risk management, including BitSight's inaugural machine learning product (patent), forecasting models, multiple analytical tools, and observational studies to evaluate internal processes.
- Supervised a Master's-level intern working on validation of external financial data and sales/marketing analytics.